



2025 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

DO NOT REMOVE PAPER FROM EXAMINATION ROOM

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Centre Number

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Student Number

Chemistry

Morning Session
Tuesday, 12 August 2025

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- Calculators approved by NESA may be used
- Use the Multiple-Choice Answer Sheet provided
- A formulae sheet, data sheet and Periodic Table are provided SEPARATELY
- Write your Centre Number and Student Number at the top of this page

Total marks:
100

Section I – 20 marks (pages 2–9)

- Attempt Questions 1–20
- Allow about 35 minutes for this section

Section II – 80 marks (pages 10–33)

- Attempt Questions 21–36
- Allow about 2 hours and 25 minutes for this section

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Section I

20 marks

Attempt Questions 1–20

Allow about 35 minutes for this part

Use the Multiple-Choice Answer Sheet for Questions 1–20.

- 1 Some compounds contain water molecules that are directly incorporated into their crystal structure, such as $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$.

What is the correct IUPAC name for $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$?

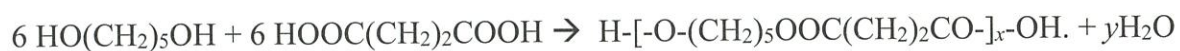
- A. Copper(I) sulfate perhydrate
 - B. Copper(I) sulfate pentahydrate
 - C. Copper(II) sulfate perhydrate
 - D. Copper(II) sulfate pentahydrate
- 2 Solid dehydrated cobalt(II) chloride is coloured blue. When it reacts with water, pink hydrated cobalt(II) chloride is produced. This is shown by the following equilibrium system.



What is the effect on the equilibrium system, and the resulting colour change, when more water is added?

- A. Equilibrium shifts to the left, becoming more blue.
 - B. Equilibrium shifts to the left, becoming more pink.
 - C. Equilibrium shifts to the right, becoming more blue.
 - D. Equilibrium shifts to the right, becoming more pink.
- 3 Which of the following correctly describes the importance of determining the concentration of heavy metal ions in soil that is used for growing food?
- A. They decrease soil pH.
 - B. They are toxic to humans.
 - C. They increase soil fertility.
 - D. They decrease soil erosion.

- 4 A part of a polymer is formed with the reactions of a pentan-1,5-diol with butanedioic acid. A section of 6 molecules of each were reacted together. The equation is shown below.



Which row of the table correctly shows the number of polymer units and the number of moles of water formed.

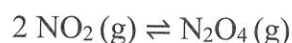
	<i>Number of polymer units (x)</i>	<i>Number of moles of water formed (y)</i>
A.	6	5
B.	6	6
C.	6	11
D.	6	12

- 5 When magnesium burns in the presence of oxygen, magnesium oxide is produced. The entropy change (ΔS) for the reaction is $-216.53 \text{ J K}^{-1} \text{ mol}^{-1}$ and the enthalpy change (ΔH) is $-601.83 \text{ kJ mol}^{-1}$.

Which of the following statements is correct?

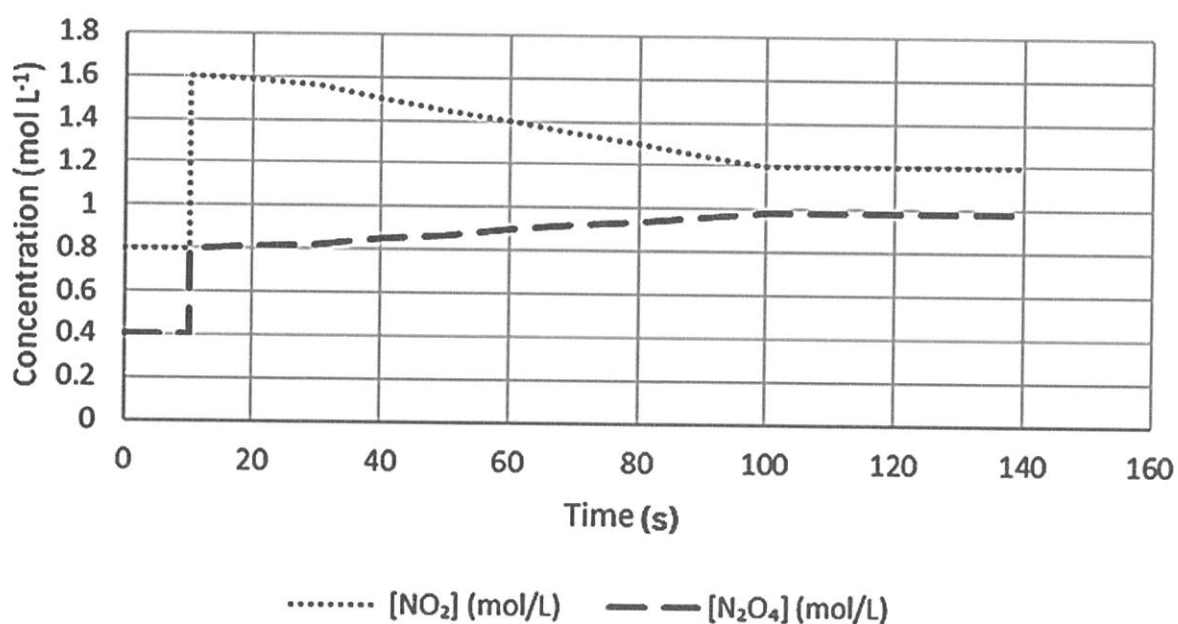
- A. The reaction is spontaneous but reversible.
 - B. The reaction is non-spontaneous and reversible.
 - C. The reaction is spontaneous and non-reversible.
 - D. The reaction is non-spontaneous and non-reversible.
- 6 What is the most appropriate method to determine the concentration of calcium in bottled water?
- A. Adding fluoride ions.
 - B. Performing a flame test.
 - C. Using mass spectroscopy.
 - D. Using atomic absorption spectroscopy.

- 7 An equilibrium was established between NO_2 and N_2O_4 represented by the equation below.



At $t = 10 \text{ s}$, a change caused the system to shift, and a new equilibrium was reached at $t = 100 \text{ s}$.

What change occurred at $t = 10 \text{ s}$ that led to the shift in equilibrium?



- A. A decrease in pressure
- B. An increase in pressure
- C. A decrease in temperature
- D. An increase in temperature
- 8 Pentanal was reacted with acidified $\text{K}_2\text{Cr}_2\text{O}_7$, and the product was immediately distilled from the reaction vessel.
- Which of the following was the product that was distilled?
- A. Pentane
- B. Pent-2-ene
- C. Pentan-1-ol
- D. Pentanoic acid

- 9 Which row of the table correctly describes the acid-base reaction of the hydrogen sulfate ion with water?

	<i>Acid</i>	<i>Base</i>	<i>Conjugate acid</i>	<i>Conjugate base</i>
A.	HSO_4^-	H_2O	SO_4^{2-}	H_3O^+
B.	H_3O^+	HSO_4^-	H_2SO_4	H_2O
C.	HSO_4^-	H_2O	H_3O^+	SO_4^{2-}
D.	H_2O	HSO_4^-	SO_4^{2-}	H_3O^+

- 10 Indigenous Australians consumed the seeds of cycad plants, which contain the toxin cycasin. The seeds were often prepared using a variety of processes to remove the toxin.

Which of the following options is the most effective process of leaching to remove the toxins?

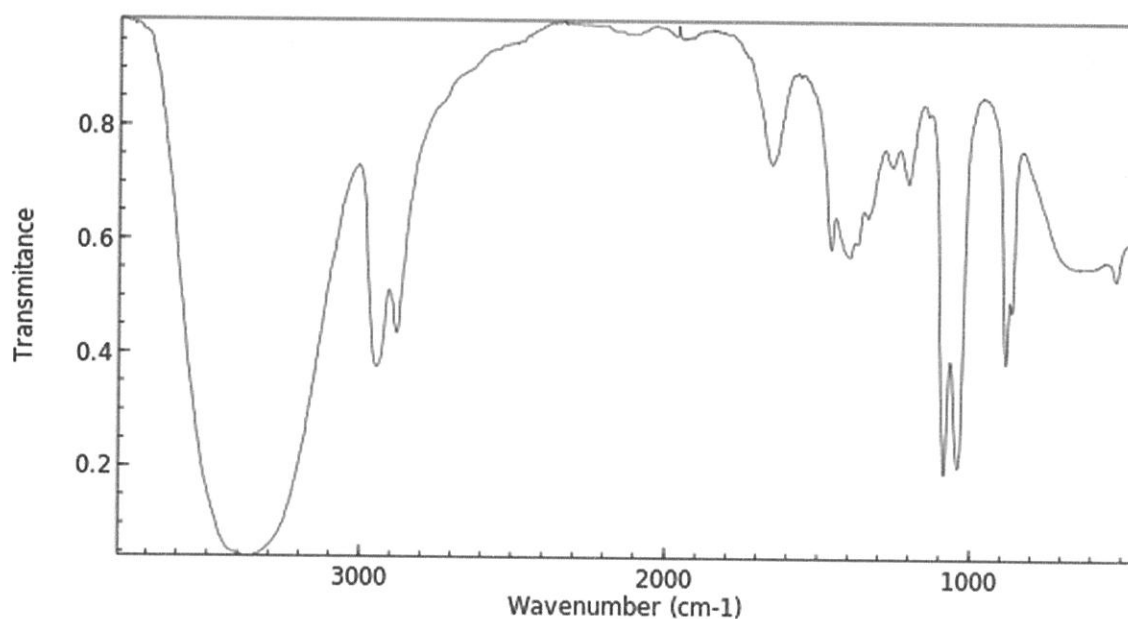
- A. Cycad kernels are placed in running water for at least three days. After this, the kernels are removed from the water, and they are cooked and eaten.
- B. Cycad kernels are placed in a still water source for at least three days. The kernels are then removed from the water to dry, before they are cooked and eaten.
- C. Cycad kernels are ground into flour and placed in a still water source overnight before they are removed from the water. The ground kernels are then dried, cooked and eaten.
- D. Cycad kernels are cut open and placed into running water for at least three days. After this, the kernels are removed from the water to dry. They are then ground, cooked and eaten.

- 11 Identify the correct equation for the complete combustion of but-2-ene.

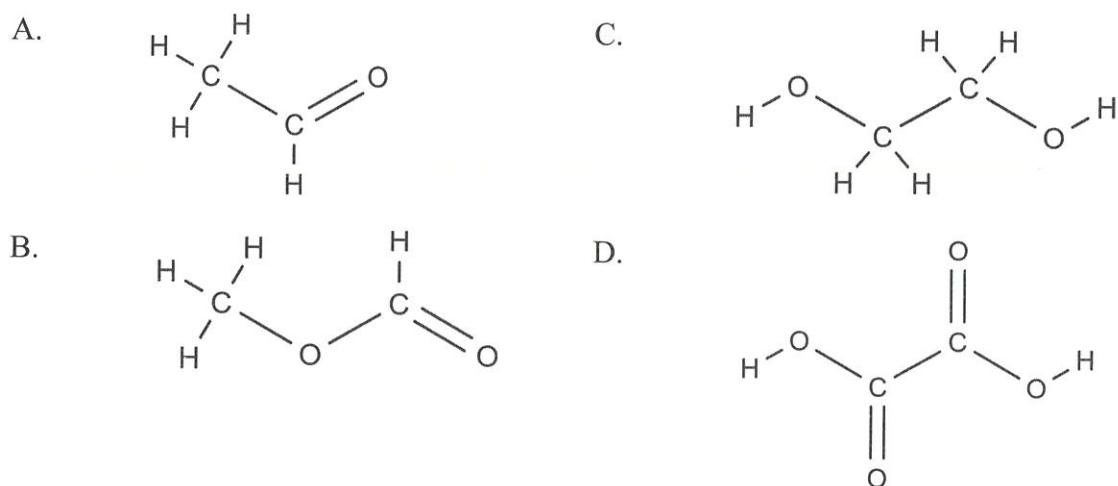
- A. $\text{C}_4\text{H}_8(\text{g}) + 6 \text{O}_2(\text{g}) \rightarrow 4 \text{CO}_2(\text{g}) + 4 \text{H}_2\text{O}(\text{g})$
- B. $\text{C}_4\text{H}_6(\text{g}) + 6 \text{O}_2(\text{g}) \rightarrow 4 \text{CO}_2(\text{g}) + 4 \text{H}_2\text{O}(\text{g})$
- C. $2 \text{C}_4\text{H}_6(\text{g}) + 11 \text{O}_2(\text{g}) \rightarrow 8 \text{CO}_2(\text{g}) + 6 \text{H}_2\text{O}(\text{g})$
- D. $2 \text{C}_4\text{H}_{10}(\text{g}) + 13 \text{O}_2(\text{g}) \rightarrow 8 \text{CO}_2(\text{g}) + 10 \text{H}_2\text{O}(\text{g})$

- 12 What colour precipitate would you expect to observe when adding a few drops of 0.1M iron(II) nitrate to a solution of 0.1M sodium hydroxide?
- Blood red
 - Light blue
 - Dark green
 - Rusty orange

- 13 This is an infrared spectrum for an organic compound:



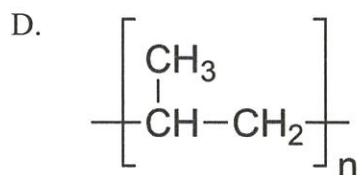
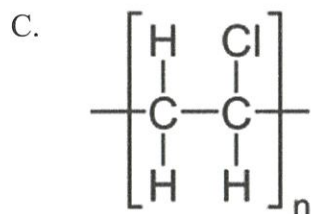
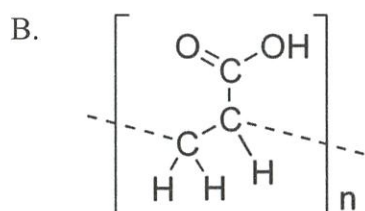
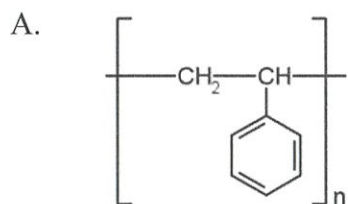
Which of the following compounds would produce the spectrum above?



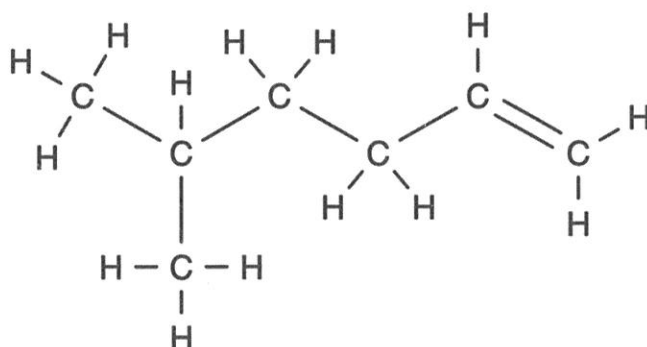
- 14 Using the table of K_b and pK_b values (at 25°C), identify the strongest base in the table.

Base	K_b	pK_b
NH_3	1.7×10^{-5}	
CH_3NH_2		9.24
CH_3COO^-	4.4×10^{-4}	
$\text{C}_5\text{H}_5\text{N}$		8.82

- A. CH_3COO^-
 B. CH_3NH_2
 C. $\text{C}_5\text{H}_5\text{N}$
 D. NH_3
- 15 Which of the following polymer structures will be the most soluble in water, if there are ten polymer units?



- 16 What is the name of the compound below, using the IUPAC naming system.



- A. 1,1-dimethylpent-4-ene
 B. 5,5-dimethylpent-1-ene
 C. 2-methylhex-5-ene
 D. 5-methylhex-1-ene
- 17 Which indicator in the table would be best for distinguishing between tomato juice (pH = 5.0) and sea water (pH 8.1)?

<i>Indicator</i>	<i>Colour below endpoint pH</i>	<i>Endpoint range</i>	<i>Colour above endpoint pH</i>
Methyl Orange	Red	pH 3.1 – 4.4	Yellow
Bromothymol Blue	Yellow	pH 6.0 – 7.6	Blue
Phenolphthalein	Colourless	pH 8.3 – 10.0	Pink
Litmus	Red	pH 4.5 – 8.3	Blue

- A. Bromothymol Blue
 B. Litmus
 C. Methyl Orange
 D. Phenolphthalein

- 18 A solution of compound X is analysed using UV-Vis spectrophotometry at a wavelength of 450 nm.

The absorbance of the solution is 0.65. The molar absorptivity of compound X is $4.5 \times 10^3 \text{ L mol}^{-1} \text{ cm}^{-1}$ and the pathlength of the cuvette is 1.0 cm.

What is the concentration of compound X in the solution?

- A. $3.2 \times 10^{-8} \text{ mol/L}$
- B. $1.4 \times 10^{-4} \text{ mol/L}$
- C. 140 mol/L
- D. 6900 mol/L

- 19 Equal volumes of two 0.01 mol L^{-1} solutions were mixed together.

Which pair of solutions will give the greatest mass of precipitate?

- A. Ca(OH)_2 and MgCl_2
- B. Ca(OH)_2 and MgSO_4
- C. Ca(OH)_2 and NaCl
- D. Ca(OH)_2 and Na_2SO_4

- 20 A buffer solution of acetic acid and sodium acetate was prepared by dissolving 0.027 moles of the solid sodium acetate salt (NaCH_3COO) in a 50 mL solution of 0.73 mol L^{-1} acetic acid (CH_3COOH). The K_a value of the reaction is 1.7×10^{-5} .

What is the pH of the buffer solution?

- A. 2.4
- B. 3.3
- C. 4.6
- D. 6.5

Section II

80 marks

Attempt Questions 21–36

Allow about 2 hours and 25 minutes for this section

-
- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
 - Show all relevant working in questions involving calculations.
 - Extra writing space is provided on page 34. If you use this space, clearly indicate which question you are answering.
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Question 21 (3 marks)

Explain why it is important to monitor the concentration of inorganic ions in the environment, using a specific NAMED example that you have studied. 3

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Question 22 (5 marks)

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A student was experimenting with the fermentation process. Twenty grams of glucose was added to 200 mL of warm water. The flask was isolated from oxygen, yeast was added and the solution was warmed to 37°C. The amount of gas produced was 3 L (25°C and 1 atm).

The resulting solution was then distilled using a new distillation process. The extracted ethanol was burnt in a bomb calorimeter with excess oxygen. The calorimeter contained 500 mL of water. The temperature was initially at 24°C and final temperature was 79.5°C. (Assume the molar heat of combustion of ethanol is 1360 kJ mol⁻¹.)

Determine the efficiency of this new distillation process.

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Question 23 (4 marks)

Water can act as an amphiprotic substance. With reference to the relevant acid-base theory, explain the amphiprotic nature of water. Include appropriate chemical equations.

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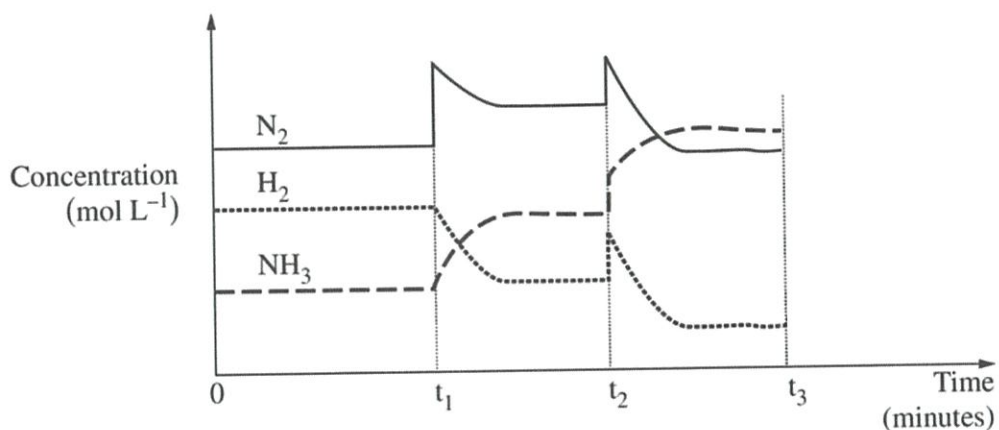
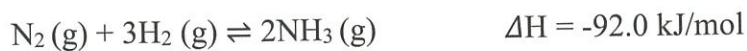
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Question 24 (5 marks)

The below graph shows the change in reaction rate with time for the following reaction.



- (a) Write the K_{eq} expression for this reaction.

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- (b) With reference to collision theory, explain the changes between t_1 and t_2 .

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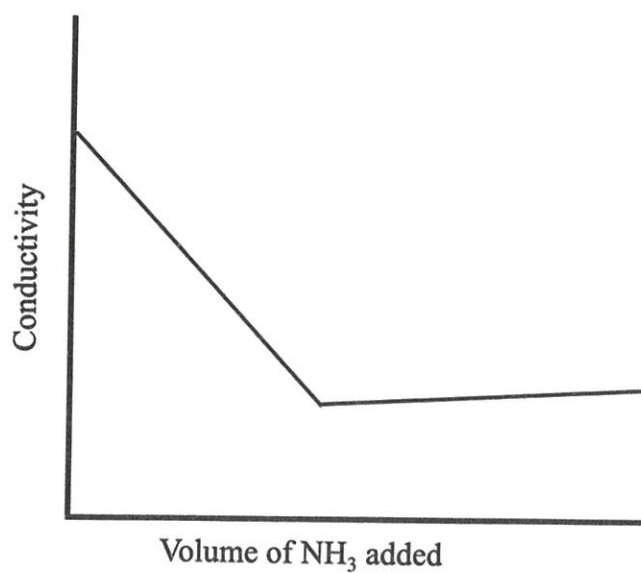
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- (c) At t_3 , the temperature of the system was increased. **On the graph above**, sketch the changes to the concentration of each chemical species.

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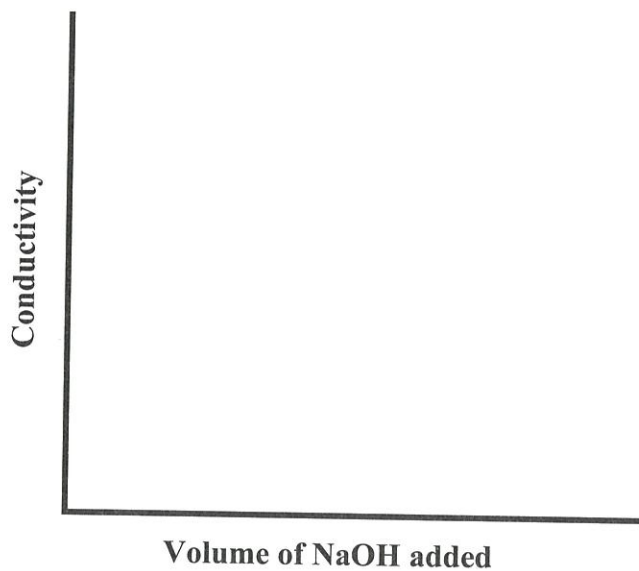
Question 25 (4 marks)

A solution of ammonia was titrated from a burette into a beaker containing hydrochloric acid. This reaction was measured using a conductivity meter and produced the graph below.



- (a) Sketch the graph if sodium hydroxide is used instead of ammonia.

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Question 25 continues on page 15

Question 25 (continued)

- (b) Account for the similarities and differences in the shapes of the two conductivity graphs. Include at least TWO relevant chemical equations. 3

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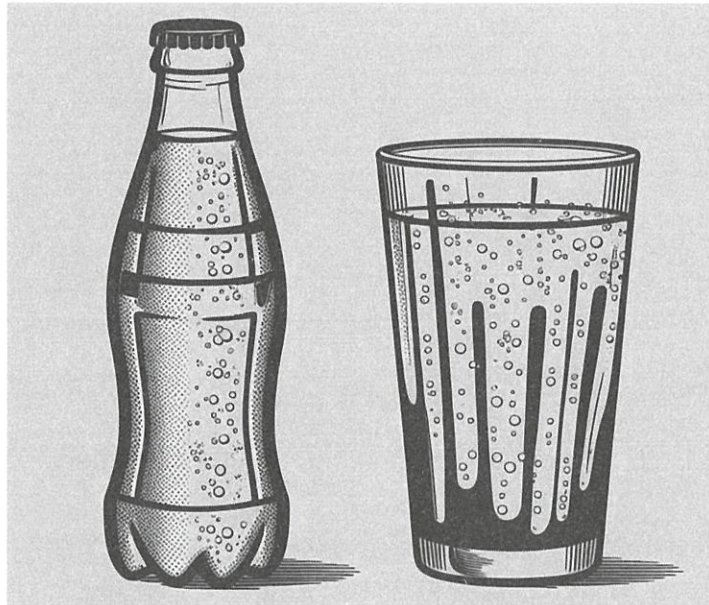
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End of Question 25

Question 26 (4 marks)

When carbon dioxide dissolves in water, carbonic acid is formed. This is known as carbonated water and commonly referred to as soda water.



- (a) Explain why dynamic equilibrium is only reached in a closed system, with reference to the stimulus above. Include a relevant chemical equation. 2

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- (b) Identify ONE benefit and ONE limitation of the model above in explaining open and closed systems, and their relationship to dynamic equilibrium. 2

ONE benefit:

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ONE limitation:

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Question 27 (4 marks)

Muntrie berries are an indigenous super food, rich in vitamin A, vitamin C, magnesium and iron. They contain more antioxidants than blueberries and have a distinct spicy apple flavour. The flavouring is due to the ester, methyl butanoate.

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A student wanted to replicate the distinct smell of Muntrie berries using school laboratory equipment.

Draw a labelled flowchart below, indicating the conditions and reactant(s) required for each step to produce a pure sample of spicy apple flavour from the starting ingredients of methanol and butanol.

Question 28 (6 marks)

As land clearing and irrigation continues, water tables continue to rise. This results in the salt level in our water supply increasing. Our drinking water is becoming harder as the concentration of magnesium ions (Mg^{2+}) and calcium ions (Ca^{2+}) increases.

- (a) Explain the effect of increased hardness on the effectiveness of soap for cleaning. 2

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- (b) *“Synthetic detergents are far more effective in hard water than soap.”* 4

Assess this statement, with reference to the structures of synthetic detergents.

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Question 28 continues on page 19

Question 28 (continued)

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End of Question 28

Question 29 (3 marks)

A small sample (roughly half a teaspoon or spatula) of white solid was found in a test tube. The student believed it was barium carbonate ($K_{sp} = 2.58 \times 10^{-9}$).

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Outline, with justification, a series of chemical tests for this small sample, that would confirm the identity as barium carbonate, given no access to a Bunsen burner or flame.

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Question 30 (5 marks)

A 0.10 mol L^{-1} solution of silver acetate was prepared. A solution of silver sulfate with unknown concentration was slowly added to 40.0 mL of the silver acetate solution of known concentration, with mixing. After 20.0 mL of the silver sulfate solution was added, a faint precipitate could be detected.

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Determine the molar solubility of the silver sulfate solution, assuming that the added moles of silver ions are negligible compared to the number of moles of silver ions in the silver acetate solution.

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Question 31 (5 marks)

A student had three samples of acids with the same concentration and tested the pH of each.

<i>Acid</i>	<i>Concentration (mol/L)</i>	<i>pH</i>
Acetic acid	0.1	3.0
Sulfuric acid	0.1	0.7
Hydrochloric acid	0.1	1.0

- (a) Account for the differences in pH values between hydrochloric acid and acetic acid. Include a relevant chemical equation. 2

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- (b) The first ionisation of sulfuric acid is almost complete, however, the second ionisation is relatively weak ($pK_a = 1.92$). Assuming 100% first ionisation, calculate the concentration of the sulfate ion for a 0.1 M solution of sulfuric acid. 3

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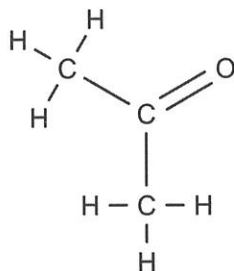
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Question 32 is on page 24

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Question 32 (6 marks)

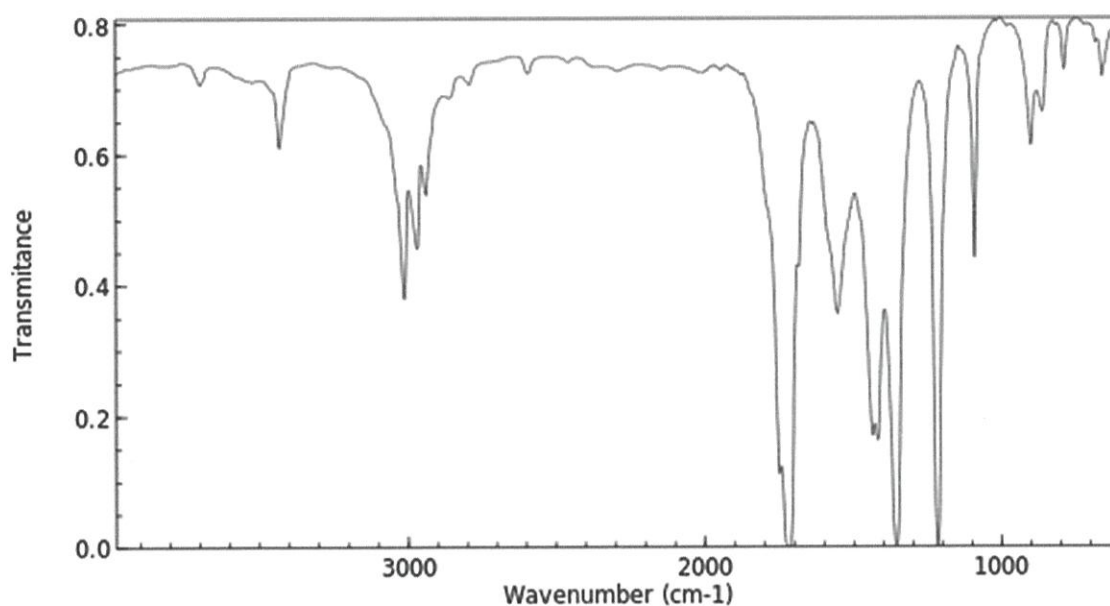
Acetone (propanone) is a common solvent used for a variety of applications including nail polish remover and as a paint thinner. Its chemical structure is shown below.



A sample of acetone is analysed using different spectroscopic techniques, including infrared, proton-NMR and carbon-13 NMR.

- (a) On the infrared spectrum below, identify the signal that is produced by the carbonyl group (C=O).

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Question 32 continues on page 25

Question 32 (continued)

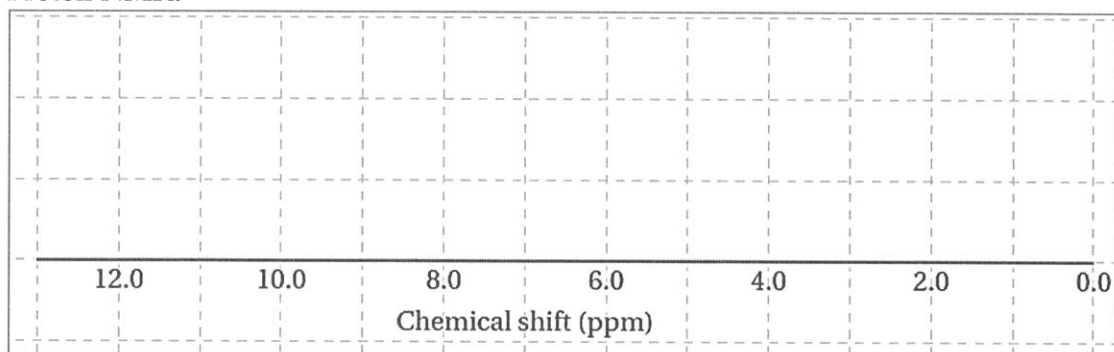
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- (b) Draw the expected signals for the proton NMR and carbon-13 NMR on the blank spectrum templates and justify your answers using the information provided below and in the data sheet.

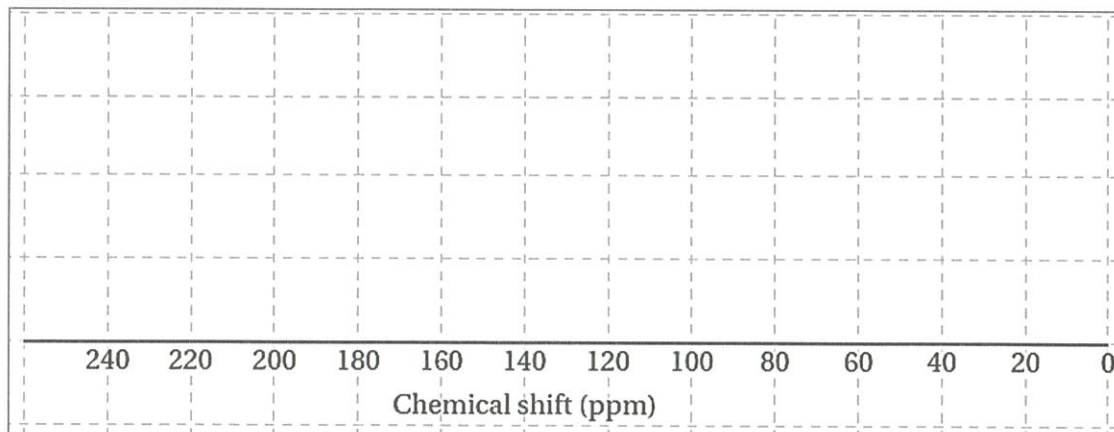
Reference proton chemical shift data

Type of proton	δ/ppm
$-\text{CH}_3$, $-\text{CH}_2-$, $-\text{CH}-$	0.7–2.1
$\left. \begin{array}{l} \text{H}_3\text{C}-\text{CO}- \\ \text{H}_2\text{C}-\text{CO}- \\ \text{H}_2\text{C}-\text{CO} \end{array} \right\}$ (aldehydes, ketones, carboxylic acids or esters)	2.2–2.6
$\left. \begin{array}{l} \text{H}_3\text{C}-\text{O}- \\ \text{H}_2\text{C}-\text{O}- \\ \text{HC}-\text{O}- \end{array} \right\}$ (alcohols or esters)	3.2–5.0

Proton NMR:



Carbon-13 NMR:



Question 32 continues on page 26

[illegible]

End of Question 32

Question 33 (4 marks)

Calculate the resultant pH of a solution prepared by combining 30.0 ml of a 0.080 mol L⁻¹ NaOH solution and 10.0 ml of a 0.010 mol L⁻¹ H₂SO₄ solution.

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Question 34 (7 marks)

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Phosphoric acid is an inorganic triprotic acid that is used as a raw material in the production of fertilisers. A sample with an unknown concentration was found and is required to be above 10 mol L^{-1} to be used in the manufacturing of fertiliser.

To analyse the phosphoric acid, 25.00 mL of the acid was diluted to 2.500 L . A 15.00 mL aliquot of the diluted solution was titrated with a standardised $0.0800 \text{ mol L}^{-1} \text{ NaOH}$ solution to the endpoint.

The following data was collected during the analysis of the phosphoric acid.

<i>Titration #</i>	<i>Volume (mL)</i>
Rough	36.30
1	35.15
2	35.10
3	35.15
4	35.10

Determine if the concentration of the phosphoric acid is suitable to be used in the manufacturing of fertiliser. Show all working.

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Question 34 continues on page 29

Question 34 (continued)

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End of Question 34

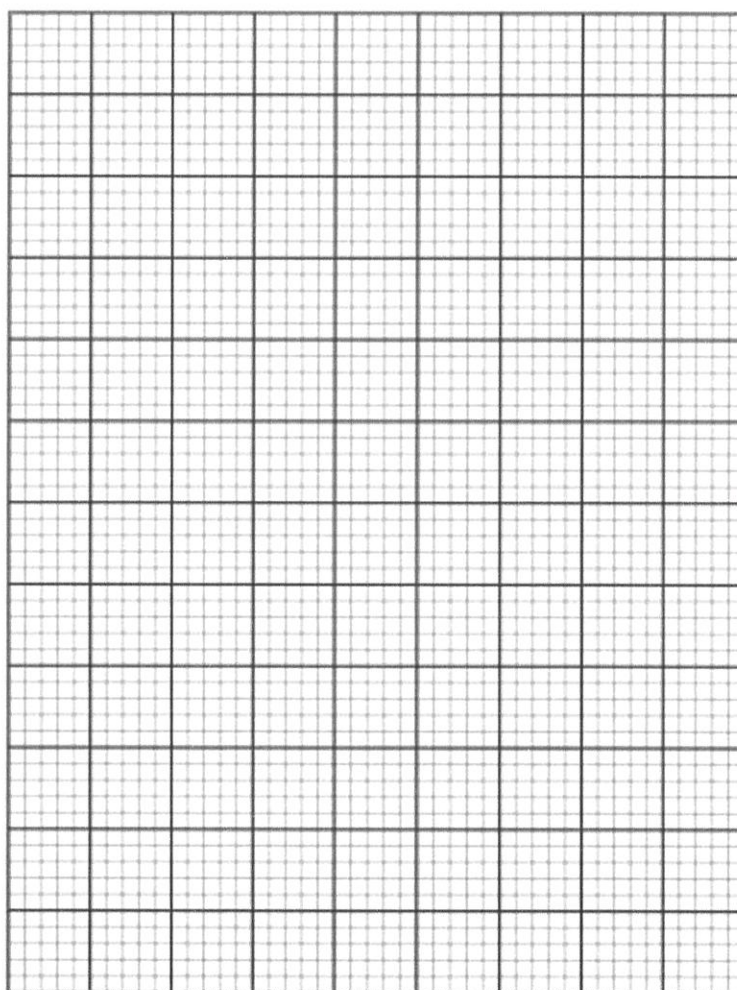
Question 35 (7 marks)

A researcher had collected the following data about two homologous series of organic compounds. Data was collected and displayed in the table below.

<i>Compound Series 1</i>	<i>Alkyl chain mass (g/mol)</i>	<i>Boiling Point (°C)</i>
C ₂ H ₄ O	28	20
C ₃ H ₆ O	42	49
C ₄ H ₈ O	58	75

<i>Compound Series 2</i>	<i>Alkyl chain mass (g/mol)</i>	<i>Boiling Point (°C)</i>
C ₂ H ₅ OH	29	78
C ₃ H ₇ OH	43	97
C ₄ H ₉ OH	57	117

- (a) Plot the boiling point as a function of alkyl chain mass for each series.

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Alkyl Chain Mass (g/mol)

Question 35 continues on page 31

Question 35 (continued)

- (b) Using the data provided, explain the similarities in the trends of the two graphs and any differences between the two graphs.

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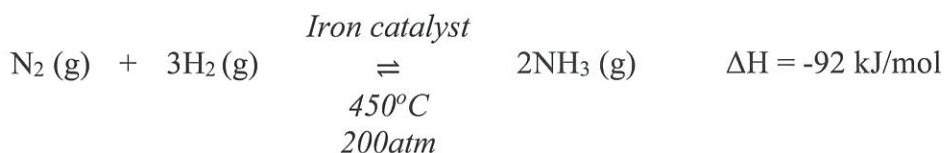
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End of Question 35

Question 36 (8 marks)

The Haber-Bosch process is the chemical synthesis reaction that produces ammonia from nitrogen and hydrogen, as summarised below.

8



First industrialised over 100 years ago, it remains a significant industrial process in the manufacture of ammonia for chemical fertilisers, explosives, plastics and many other products. However, the Haber-Bosch process has very large energy demands and it is estimated that the process accounts for 1–2% of global energy consumption and 3% of carbon emissions.

Evaluate the environmental and economic impacts of the successful development of a new catalyst that enables ammonia production at a much lower temperature in the Haber-Bosch process.

Refer to the information provided and your understanding of relevant chemical principles in your response.

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Question 36 continues on page 33

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End of Examination

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Copyright

Question	Citation
13	Spectrum comes from <i>sdb.sdb.aist.go.jp</i> . (n.d.). <i>SDBS, Spectral Database for Organic Compounds, AIST</i> . [online] Available at: https://sdb.sdb.aist.go.jp/ . [Accessed 18 Mar. 2025]. Structures drawn by Elizabeth Cranfield – 2025 CSSA Trial HSC Examinations Chemistry – Committee. MolView (https://molview.org/).
15	Option A: <i>Wikimedia.org</i> . (2022). <i>File:Polystyrene PS.png - Wikimedia Commons</i> . [online] Available at: https://commons.wikimedia.org/wiki/File:Polystyrene_PS.png [Accessed 18 Mar. 2025]. Option B: <i>Wikimedia.org</i> . (2009). <i>File:Polyacrylic acid.svg - Wikimedia Commons</i> . [online] Available at: https://commons.wikimedia.org/wiki/File:Polyacrylic_acid.svg [Accessed 18 Mar. 2025]. Option C: <i>Wikimedia.org</i> . (2008). <i>File:Polyvinylchlorid.svg - Wikimedia Commons</i> . [online] Available at: https://commons.wikimedia.org/wiki/File:Polyvinylchlorid.svg . [Accessed 18 Mar. 2025]. Option D: Ed. (2011, October 31). <i>English: Chemical structure of polypropylene</i> . Wikimedia Commons. https://commons.wikimedia.org/wiki/File:Polypropylene.svg [Accessed 18 Mar. 2025].
16	Structure drawn by Scott Tibbey - 2025 CSSA Trial HSC Examinations Chemistry – Committee. MolView (https://molview.org/).
32	Structures drawn by Elizabeth Cranfield - 2025 CSSA Trial HSC Examinations Chemistry – Committee. MolView (https://molview.org/).
32(a)	Spectrum comes from <i>sdb.sdb.aist.go.jp</i> . (n.d.). <i>SDBS, Spectral Database for Organic Compounds, AIST</i> . [online] Available at: https://sdb.sdb.aist.go.jp/ . [Accessed 18 Mar. 2025].

EXAMINERS

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Chemistry

Section I

20 marks

Questions 1–20 (1 mark each)

Question	Answer	Outcomes Assessed	Targeted Performance Band
1	D	CH12-15	2–3
2	D	CH11/12-5, CH12-12	2–3
3	B	CH12-15	2–3
4	A	CH12-14	2–3
5	C	CH12-12	3–4
6	D	CH11/12-2, CH12-15	3–4
7	B	CH11/12-5, CH12-12	3–4
8	D	CH12-14	3–4
9	C	CH12-13	3–4
10	D	CH11/12-2, CH12-12	3–4
11	A	CH12-14	3–4
12	C	CH12-15	3–4
13	C	CH11/12-5, CH12-15	4–5
14	A	CH11/12-5, CH12-13	4–5
15	B	CH12-14	4–5
16	D	CH12-14	4–5
17	A	CH11/12-5, CH11/12-6, CH12-13	4–5
18	B	CH12-15	5–6
19	A	CH11/12-5, CH12-12, CH12-15	5–6
20	C	CH11/12-5, CH12-13	5–6

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Section II

80 marks

Question 21 (3 marks)

Outcomes Assessed: CH11/12-7, CH12-15

Targeted Performance Bands: 2–4

Criteria	Marks
• Explains the importance of monitoring a specific inorganic ion in the environment	3
• Outlines the importance of monitoring the environment OR • Explains the importance of monitoring organic or molecular species in the environment e.g. CO ₂	2
• Provides relevant information	1

Sample Answer:

There are many possibilities. Here are two examples of ions.

Lead (Pb²⁺) is toxic to humans and accumulates in the body tissue if it is consumed. It is important that soil used for growing food crops is tested for lead before consumption, as plants absorb lead from the soil and would contaminate the food that the humans eat.

Phosphates (PO₄³⁻) are a nutrient for plants and are in fertilisers. When fertiliser containing phosphates runs into bodies of water, it disrupts the ecosystem, as it promotes the growth of algae and causes eutrophication. This leads to death of fish and other aquatic organisms and the algae may produce toxins that are harmful to human health.

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Question 22 (5 marks)**Outcomes Assessed: CH11/12-5, CH12-14****Targeted Performance Bands: 2–6**

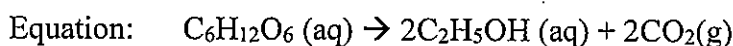
Criteria	Marks
- Determines the moles of ethanol and carbon dioxide produced - Provides a correct fermentation equation - Calculates the moles of ethanol from the heat of combustion - Calculates correct answer for efficiency percentage - Provides the final answer to the correct number of significant figures	5
Provides substantially correct steps in the calculation, but with an error	4
Provides some correct steps in the calculation	3
Provides ONE correct step in the calculation	2
Provides any relevant information	1

Sample Answer:

$$MM(\text{glucose}) = 180.156 \text{ g/mol}; MM(\text{ethanol}) = 46.068 \text{ g/mol}$$

$$V_m(\text{at } 25^\circ\text{C}) = 24.79 \text{ L}$$

$$n(\text{glucose}) = 20/180.156 = 0.111 \text{ mol}$$



$$n(CO_2) = 3/24.79 = 0.1210 \text{ mol}$$

$$m(\text{ethanol}) = 0.1210 \times 46.068 = 5.575 \text{ g}$$

$$\text{Heat of combustion. } \Delta H = mC\Delta T$$

$$500 \times 4.18 \times 55.5 = 115995 \text{ J}$$

$$n(\text{ethanol extracted}) = 115.995/1360 = 0.08529 \text{ mol}$$

$$\% \text{ efficiency} = (0.08529/0.1210) \times 100\% = 70.5\%$$

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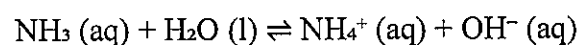
Question 23 (4 marks)**Outcome Assessed: CH12-13****Targeted Performance Bands: 2–5**

Criteria	Marks
• Explanation of an amphiprotic species AND • Explanation of Bronsted-Lowry theory of acids and bases AND • TWO equations showing the amphiprotic nature of water	4
• Explanation of an amphiprotic species AND • Identification of Bronsted-Lowry theory of acids and bases AND • ONE relevant equation showing water acting as an acid OR a base	3
• Explanation of an amphiprotic species OR • Identification of Bronsted-Lowry theory of acids and bases; AND • ONE relevant equation showing water acting as an acid OR a base	2
• Any relevant information	1

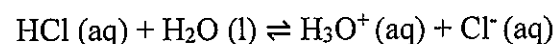
Sample Answer:

Water is considered an amphiprotic substance because it can act both as an acid and as a base. This property can be explained using the Brønsted-Lowry theory, which defines acids as proton donors and bases as proton acceptors.

Water can donate a proton (H^+) to a base. For example, in the reaction with ammonia:



Water can also accept a proton from an acid. For instance, in the reaction with hydrochloric acid:

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Question 24 (5 marks)**Question 24 (a) (1 mark)****Outcomes Assessed:** CH11/12-5, CH12-12**Targeted Performance Bands:** 3–4

Criteria	Marks
Correct K_{eq} expression	1

Sample Answer:

$$\frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$$

Question 24 (b) (3 marks)**Outcomes Assessed:** CH11/12-5, CH12-12**Targeted Performance Bands:** 2–5

Criteria	Marks
- Explanation of how an increase in the concentration of N_2 affects the shift in equilibrium; AND - links the increase of a species with the increase in collisions that will favour the forward reaction	3
- Explanation of how an increase in the concentration of one species affects the shift in equilibrium; OR - links the increase of a species with the increase in collisions that will favour the forward reaction	2
Any relevant information	1

Sample Answer:

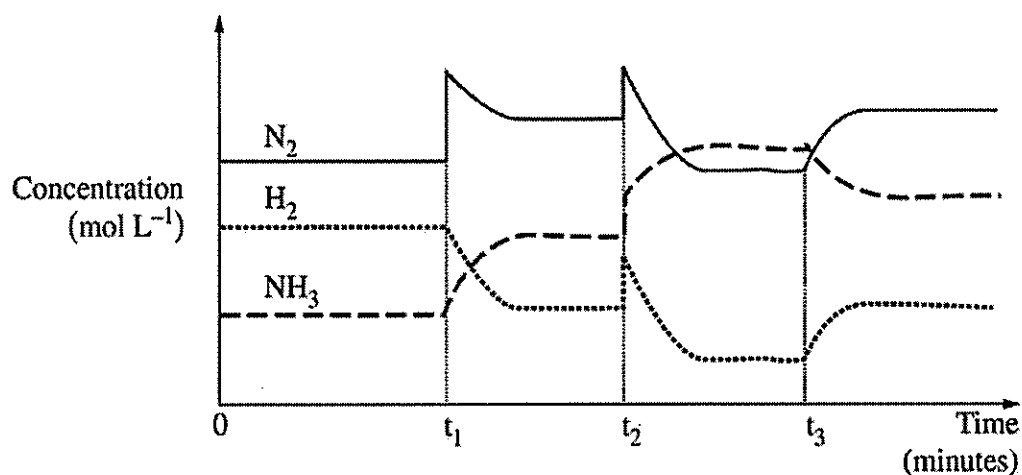
At t_1 there is an increase in the concentration of N_2 . An increase in the concentration of one species in a given space means that there is a higher chance of collisions between reactant molecules. This favours the forward reaction and leads to an equilibrium shift to the right and an increased number of NH_3 molecules being formed, until equilibrium is re-established.

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Question 24 (c) (1 mark)**Outcomes Assessed:** CH11/12-6, CH12-12**Targeted Performance Bands:** 4–5

Criteria	Marks
• Completion of the graph showing a curved rise in N_2 and H_2 and a curved fall in NH_3 (no need for the ratio of the final changes to be stoichiometrically correct)	1

Sample Answer:**Disclaimer**

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Question 25 (4 marks)

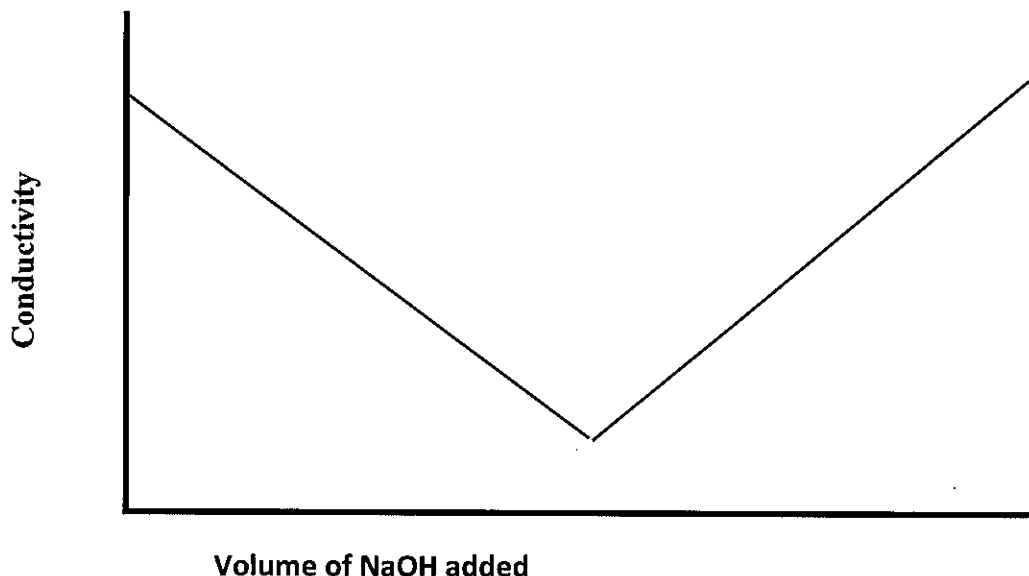
Question 25 (a) (1 mark)

Outcomes Assessed: CH11/12-6, CH12-13

Targeted Performance Bands: 3–4

Criteria	Marks
• Sketch of the conductivity graph with correct features	1

Sample Answer:



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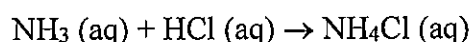
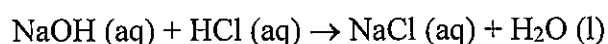
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Question 25 (b) (3 marks)**Outcomes Assessed: CH11/12-7, CH12-13****Targeted Performance Bands: 3–5**

Criteria	Marks
- Explains why the shape of the conductivity graph is the same for both graphs as the base is added until the equivalence point AND - Explains why the shape of the conductivity graph is different for both graphs after the equivalence point AND - Includes at least TWO relevant chemical equations, including the ammonia equilibrium	3
- Identifies the shape of the conductivity graph is the same for both graphs as the base is added until the equivalence point OR - Identifies the shape of the conductivity graph is different for both graphs after the equivalence point AND - Includes ONE relevant chemical equations	2
Any relevant information	1

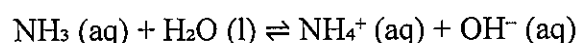
Sample Answer:

There is high initial conductivity due to H^+ ions from the HCl which fully dissociates as a strong acid. The conductivity begins to decrease as both bases are added and reacts to neutralise the H^+ ions. The equivalence point occurs at the point on the graph with the lowest conductivity.



As more NaOH is added, the conductivity increases again as it is a strong base and is fully dissociated, giving up all OH^- ions.

As more NH_3 is added, the conductivity only slightly increased as it is a weak base and only partially ionises to produce OH^- ions.

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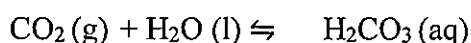
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Question 26 (4 marks)**Question 26 (a) (2 marks)****Outcomes Assessed:** CH12-12**Targeted Performance Bands:** 2–4

Criteria	Marks
Explains a requirement for a closed system to support dynamic equilibrium and states the chemical equation for the reaction	2
Includes relevant information	1

Sample Answer:

Dynamic equilibrium can only be achieved in a closed system where no matter is lost to the environment. The stimulus above shows an open system of a glass and closed system of a bottle, where dynamic equilibrium is achieved in the bottle.

**Question 26 (b) (2 marks)****Outcomes Assessed:** CH11/12-7, CH12-12**Targeted Performance Bands:** 3–5

Criteria	Marks
- Identifies ONE correct limitation AND - Identifies ONE benefit of the model	2
- Identifies ONE correct limitation OR - Identifies ONE benefit of the model	1

Sample Answer:

A benefit of the model is that it assists with understanding of an open and closed system, as it shows a glass as an open system losing matter to the environment and the bottle as a closed system.

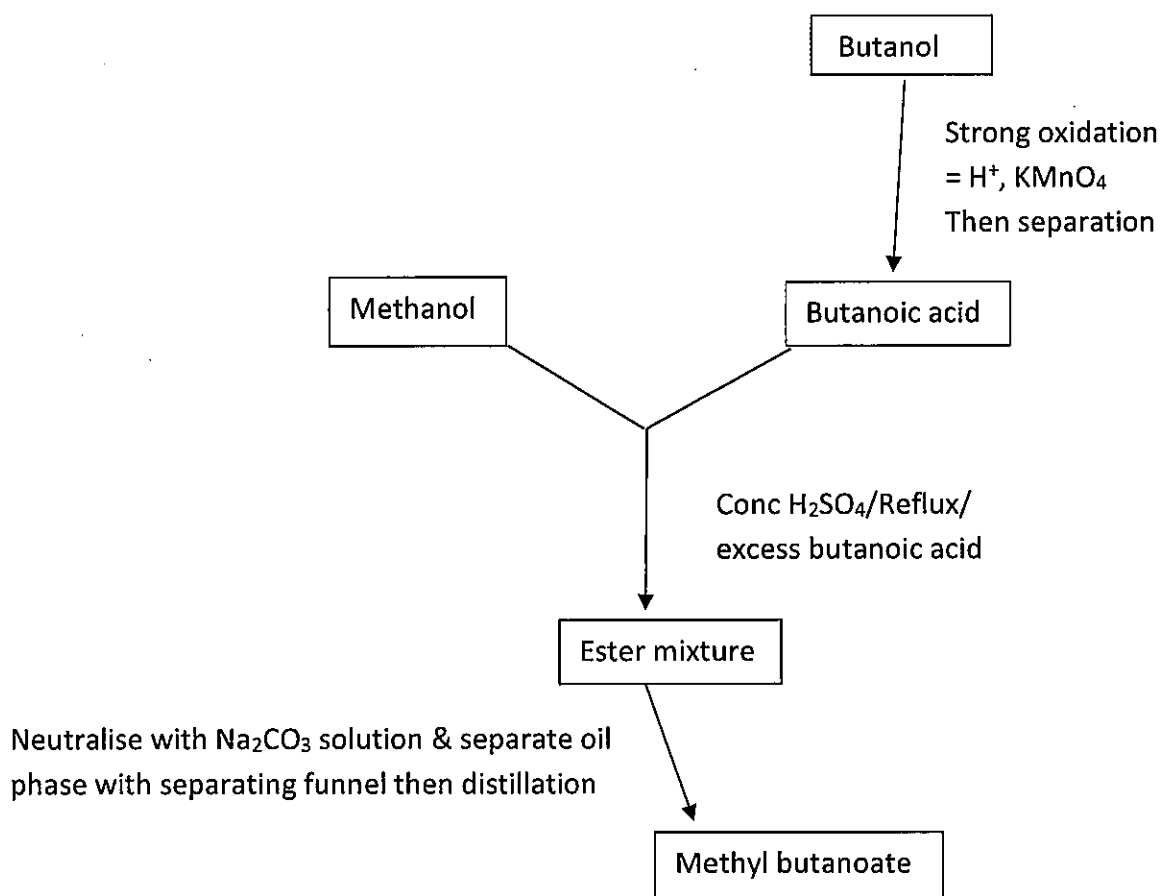
A limitation of the model is that it does not clearly demonstrate a dynamic equilibrium taking place between the gaseous carbon dioxide and the aqueous carbonic acid.

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Question 27 (4 marks)**Outcomes Assessed: CH11/12-6, CH11/12-7, CH12-14****Targeted Performance Bands: 2–6**

Criteria	Marks
• Clear flowchart • All initial and final compounds • All reaction conditions and all physical separation steps, including the distillation to extract pure methyl butanoate	4
• Clear flowchart • All initial and final compounds • Most reaction conditions and physical separation steps	3
• Two steps with initial and final compounds, with some reaction conditions or physical separation steps	2
• Identifies a step	1

Sample Answer:**Disclaimer**

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Question 28 (6 marks)**Question 28 (a) (2 marks)****Outcomes Assessed: CH12-14****Targeted Performance Bands: 2–4**

Criteria	Marks
- Describes structure of soap - Identifies the effect of magnesium and calcium ions on functional group of soap	2
Any relevant information	1

Sample Answer:

The structure of soap is a negatively charged head group connected to a long oily hydrocarbon chain.



The head group reacts with the magnesium and calcium ions to form a precipitate, known as soap scum. This precipitation reaction reduces the effectiveness of soap in hard water.

Question 28 (b) (4 marks)**Outcomes Assessed: CH12-14****Targeted Performance Bands: 3–6**

Criteria	Marks
- Outlines the structure of anionic, cationic and non-ionic detergents - Describes the effect of magnesium and calcium ions on all THREE types of detergents - Assesses the relative effects of hard water on synthetic detergents compared to soap - Makes a suitable judgement about the statement	4
- Outlines the structure of anionic, cationic and non-ionic detergents AND/OR - Describes the effect of magnesium and calcium ions on different types of detergents AND/OR - Assesses the relative effects of hard water on synthetic detergents compared to soap	2–3
Any relevant information	1

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Sample Answer:

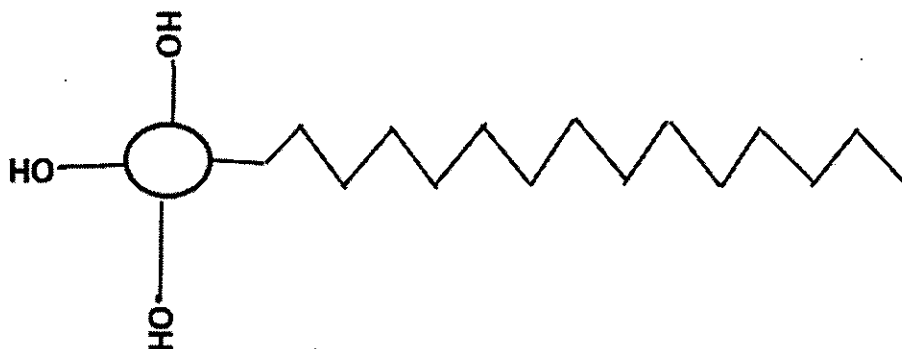
The issue with using soap in hard water is the reaction between the negative headgroup and the magnesium and calcium ions. This is still a problem for anionic detergents, which have a similar structure. However, the effect is less than for soap.



Cationic detergents have a positive headgroup, so are far more effective than anionic detergents. There is no precipitation reaction between these detergents and magnesium or calcium ions, so they are effective in hard water.



Non-ionic detergents do not have a charged head group but contain polar groups that allow for solubility in water. Like with cationic detergents, there is no precipitation reaction between these detergents and magnesium or calcium ions, so they are effective in hard water.



So cationic and non-ionic detergents are far more effective in hard water than soap, however, anionic detergents are only slightly more effective than soap.

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Question 29 (3 marks)**Outcomes Assessed:** CH11/12-2, CH11/12-6, CH12-15**Targeted Performance Bands:** 3–6

Criteria	Marks
• Justifies a series of appropriate chemical tests for the sample	3
• Outlines a suitable chemical test for barium ions AND carbonate ions	2
• Any relevant information	1

Sample Answer:

Barium carbonate has a low solubility, so dissolving in water is not appropriate.

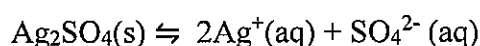
1. Add 1.0 M HCl or HNO₃ to the solid. The appearance of rapid bubbling would indicate the presence of CARBONATE ions. Add enough acid to completely dissolve the solid.
2. To the resulting solution add a few drops of 0.1 M H₂SO₄ (or another source of sulfate ions). The appearance of a milky precipitate (barium sulfate), indicates the presence of BARIUM ions.

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Question 30 (5 marks)**Outcomes Assessed:** CH11/12-5, CH11/12-6, CH12-12**Targeted Performance Bands:** 2–6

Criteria	Marks
- Correctly determines the molar solubility and shows all working. Includes the following : - Correct dissociation ionic equation K_{sp} expression - Calculates correct equilibrium values of reagents - Answer includes correct units	5
Determines the molar solubility and shows working. Includes a minor error in working listed above.	4
Attempts to calculate the final molar solubility but has multiple errors in working listed above.	3
Provides correct dissociation equation and K_{sp} expression	2
Provides some relevant information	1

Sample Answer:

$$K_{sp} = [\text{Ag}^+]^2[\text{SO}_4^{2-}]$$

Let x = number of moles of SO_4^{2-} ions added.

	$\text{Ag}_2\text{SO}_4(\text{s})$	$\rightleftharpoons$	2Ag^+	+	SO_4^{2-}
R			2		1
I			0.10×0.04		0
C			$+2x$		$+x$
E			$0.10 \times 0.04 + 2x$		x

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For $n(\text{Ag}^+) = 0.10 \times 0.04 + 2x$, assume $2x$ is negligible, so $n(\text{Ag}^+) = 0.10 \times 0.04$ moles

$$[\text{Ag}^+] = n/V = 0.10 \times 0.04/0.06 = 0.06667 \text{ mol/L}$$

$$[\text{SO}_4^{2-}] = x/0.06$$

$$K_{sp} = [\text{Ag}^+]^2[\text{SO}_4^{2-}]$$

$$1.20 \times 10^{-5} = [0.06667]^2 \times x/0.06$$

$$x = 1.20 \times 10^{-5} \times 0.06/[0.06667]^2$$

$$x = 1.62 \times 10^{-4} \text{ mol}$$

Original concentration of unknown Ag_2SO_4 solution = $1.62 \times 10^{-4}/0.02 = 8.1 \times 10^{-3} \text{ mol/L}$.

Question 31 (5 marks)

Question 31 (a) (2 marks)

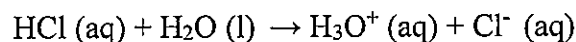
Outcomes Assessed: CH12-13

Targeted Performance Bands: 2–4

Criteria	Marks
Describes the ionisation for both weak and strong acids	2
Includes ONE relevant equation	
Any relevant information	1

Sample Answer:

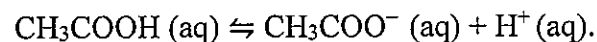
The pH is a measure of hydronium (hydrogen) ion concentration, and this is dependent on how the acid ionises. Hydrochloric acid completely ionised as it is a strong acid:



To calculate the pH of the HCl solution,

$$\text{pH} = -\log [\text{H}^+] = -\log [\text{HCl}] = -\log (0.1) = 1$$

Acetic acid is a weak acid and will only partially ionise:



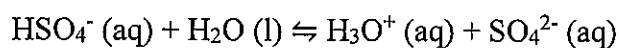
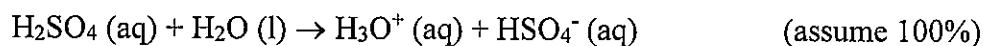
Hence the $[\text{H}^+]$ is less than 0.1 mol/L, so the pH is higher than 1.

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Question 31 (b) (3 marks)**Outcomes Assessed: CH11/12-5, CH12-13****Targeted Performance Bands: 4–6**

Criteria	Marks
• Correct expression for K for second ionisation • Correct calculation of K from pKa • Correct calculation of [H⁺] from pH • Correct calculation of [SO₄²⁻]	3
• Some correct steps in the calculation	2
• One correct step in the calculation OR a correct equation	1

Sample Answer:

$$K = \frac{[\text{SO}_4^{2-}][\text{H}_3\text{O}^+]}{[\text{HSO}_4^-]}$$

$$[\text{SO}_4^{2-}] = (K \times [\text{HSO}_4^-]) / [\text{H}_3\text{O}^+]$$

$$= (10^{-1.92} \times 0.1) / (10^{-0.7})$$

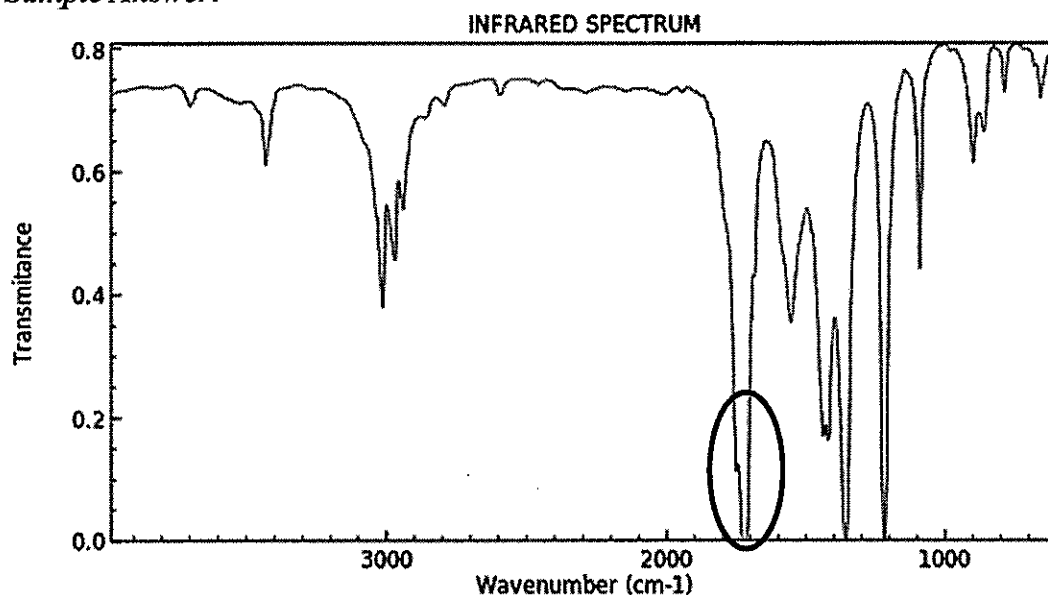
$$= 6 \times 10^{-3} \text{ mol/L}$$

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Question 32 (6 marks)**Question 32 (a) (1 mark)****Outcomes Assessed:** CH11/12-5, CH12-15**Targeted Performance Bands:** 3–4

Criteria	Marks
Identifies the correct peak	1

Sample Answer:**Question 32 (b) (5 marks)****Outcomes Assessed:** CH11/12-1, CH11/12-5, CH11/12-6, CH11/12-7, CH12-15**Targeted Performance Bands:** 2–6

Criteria	Marks
- Draws correct number, approximate location and intensity of the signals on the proton and C-13 NMR spectra - Justifies the signals using data provided/on data sheet showing extensive understanding of the interpretation of the structure	5
- Draws correct number and approximate location of the signals on the proton and C-13 NMR spectra - Justifies the signals using data provided/on data sheet showing thorough understanding of the interpretation of the structure	4
- Draws correct number of signals on the proton OR C-13 NMR spectra - Justifies the signals showing sound understanding of the interpretation of the structure	3
Draws correct number of signals on the proton OR C-13 NMR spectra	2
Provides some relevant information	1

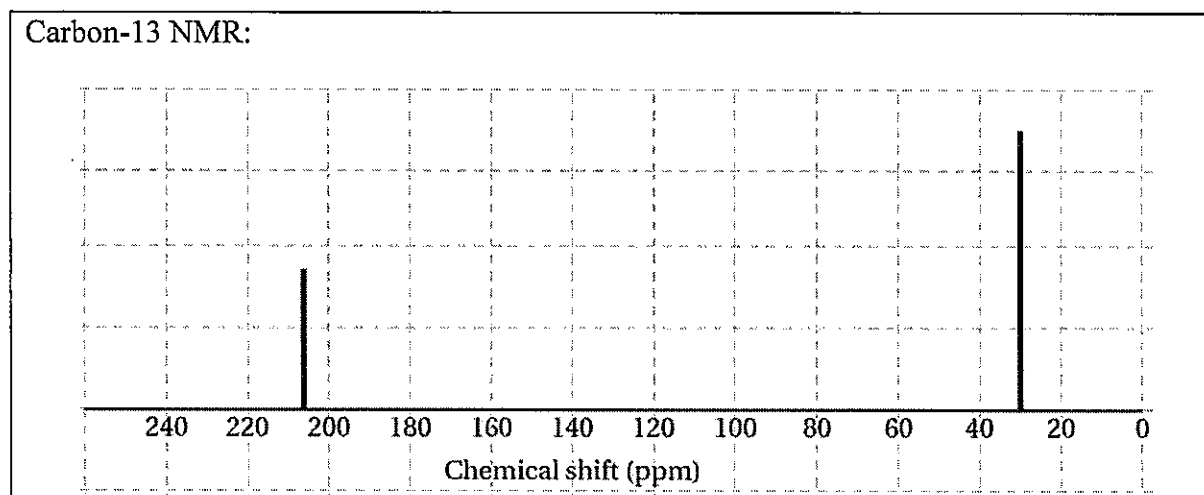
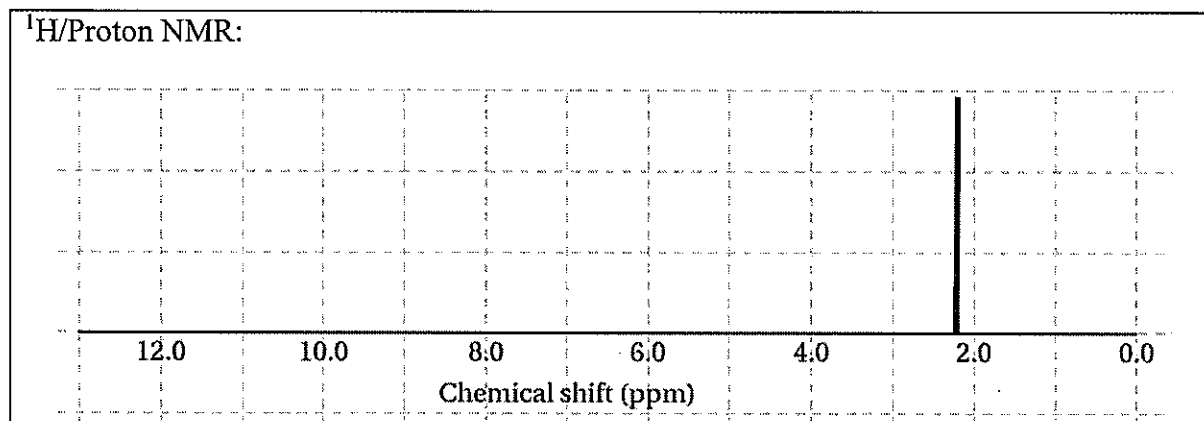
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Sample Answer:

The molecule is symmetrical and there are no hydrogens on the central carbon. Therefore, there is only 1 hydrogen environment, where all 6 hydrogens exist – 3 in each methyl group. Due to there being no hydrogens on the central carbon, there will only be a singlet. Since the hydrogens are all on a carbon that is bonded as a C=O in a ketone, the signal will be between 2.2 and 2.6 ppm. There are 2 carbon environments, again due to symmetry. The C=O will give a signal at 190-220 ppm and the 2 CH₃ groups will give a signal at 20-50 ppm, as they are bonded to a C=O. This signal will be more intense due to there being two carbons in this environment.

Note: there is no requirement for students to *explain* the shifts in terms of deshielding etc.



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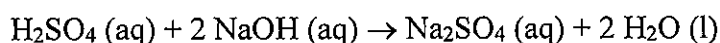
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Question 33 (4 marks)**Outcomes Assessed: CH11/12-5, CH12-13****Targeted Performance Bands: 2–6**

Criteria	Marks
• Correctly calculates the resultant pH	4
• Provides substantially correct calculations	3
• Provides some relevant steps in the calculation	2
• Provides some relevant information or working	1

Sample Answer:

Balanced Chemical Equation

**H₂SO₄**

$$n = c \times V$$

$$n = 0.010 \times 0.0100$$

$$n = 1.0 \times 10^{-4} \text{ moles } (=0.10 \times 10^{-3})$$

NaOH

$$n = c \times V$$

$$n = 0.080 \times 0.0300$$

$$n = 2.4 \times 10^{-3} \text{ moles}$$

From the equation, 1:2 mole ratio.

H₂SO₄ is the limiting reagent.NaOH is in excess by $(2.4 - 2 \times 0.1) \times 10^{-3} \text{ moles} = 2.2 \times 10^{-3} \text{ moles}$

Concentration of excess NaOH:

$$[\text{OH}^-] = n/V = 2.2 \times 10^{-3} \text{ mol}/0.04 \text{ L} = 5.5 \times 10^{-2} \text{ mol/L}$$

$$\text{pOH} = -\log [\text{OH}^-] = -\log (5.5 \times 10^{-2}) = 1.26$$

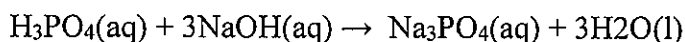
$$\text{pH} = 14 - \text{pOH} = 14 - 1.26 = 12.74$$

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Question 34 (7 marks)**Outcomes Assessed: CH11/12-5, CH11/12-6, CH12-13****Targeted Performance Bands: 2–6**

Criteria	Marks
- Calculates correct concentration of phosphoric acid to determine if it meets industry standards and all working shown. Includes the following: - Balanced chemical equation - Calculated moles of known - Calculated concentration of diluted unknown - Calculated concentration of original unknown - Correct number of significant figures (3sf) - Determines if suitable for fertiliser production	7
• A substantially correct calculation with one or two errors	5–6
• Contains two or more correct steps	3–4
• Contains ONE correct step in the calculation	2
• Provides some relevant information	1

Sample Answer:

$$\text{Average titre} = (2 \times 35.10 + 2 \times 35.15)/4 = 35.125 \text{ mL}$$

$$n(\text{NaOH}) = CV = 0.080 \times 0.035125 = 2.81 \times 10^{-3} \text{ moles}$$

$$\text{From the equation, } n(\text{NaOH}) : n(\text{H}_3\text{PO}_4) = 3:1$$

$$n(\text{H}_3\text{PO}_4) = n(\text{NaOH})/3 = 9.367 \times 10^{-4} \text{ moles}$$

$$C(\text{H}_3\text{PO}_4) = n/v = 9.367 \times 10^{-4}/0.015 = 0.062444 \text{ mol L}^{-1} \text{ (diluted)}$$

Account for dilution 25 mL:2500 mL (1:100)

$$C_1V_1 = C_2V_2 \quad (\text{OR} \quad C(\text{H}_3\text{PO}_4) \times 100)$$

$$C_1 \times 25 = 0.062444 \times 2500$$

$$= 6.2444 \text{ mol L}^{-1} \text{ (Original concentration of H}_3\text{PO}_4\text{)}$$

$$= 6.24 \text{ mol L}^{-1} \text{ (3sf)}$$

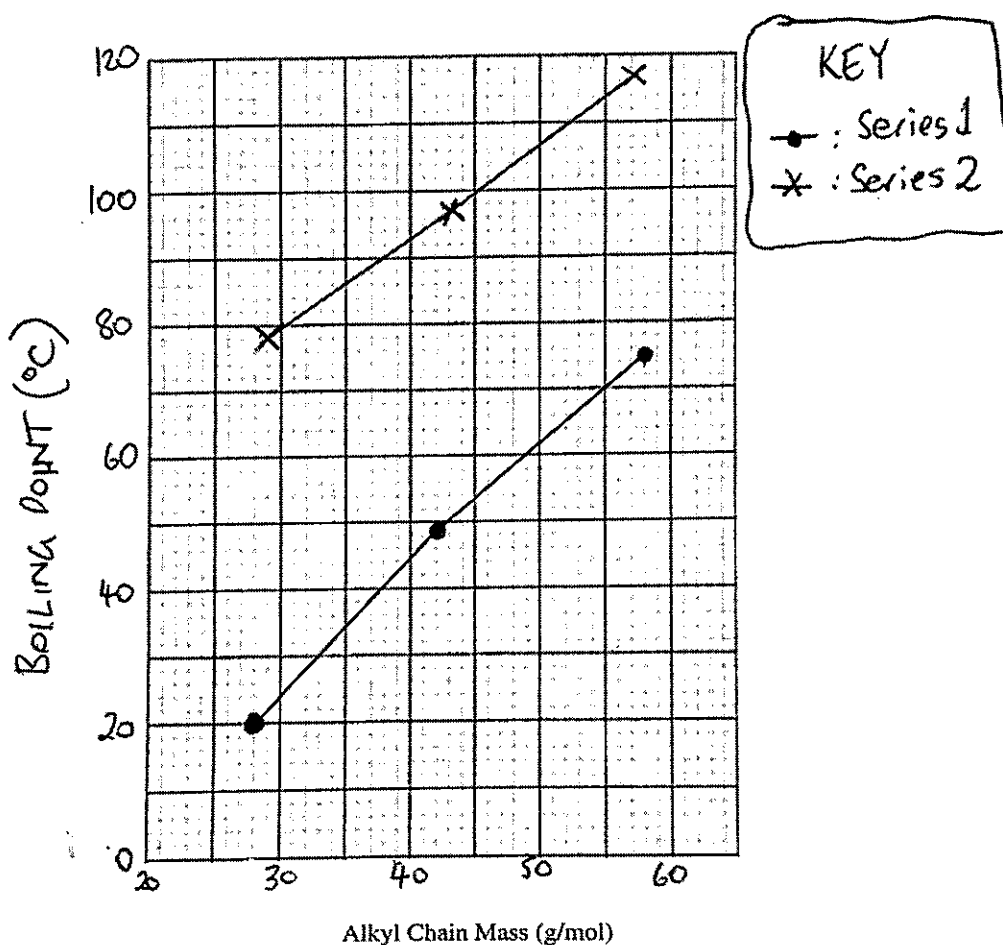
Therefore, 6.24 mol L⁻¹ of H₃PO₄ is below what is required for fertiliser production.

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Question 35 (7 marks)**Question 35 (a) (3 marks)****Outcomes Assessed:** CH11/12-5, CH12-14**Targeted Performance Bands:** 2–4

Criteria	Marks
• Correct scale and correct axes placement • Correct plotting of points • Suitable key • Labelled axes with units	3
• Correct plotting of points • Correct scale and correct axes placement	2
• ONE correct feature of the graph	1

Sample Answer:**Disclaimer**

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Question 35 (b) (4 marks)**Outcomes Assessed: CH11/12-7, CH12-14****Targeted Performance Bands: 3–6**

Criteria	Marks
- Identifies and explains differences as chain mass length increases - Identifies dipole in Series 1 and H-bonding in Series 2 - Explains the double bonded oxygen is a strong dipole - Explains H-bonding, and identifies the relative strength of H-bonding - Quotes data from table for both chain and functional group	4
- Identifies and explains differences as chain mass length increases - Identifies dipole in Series 1 and H-bonding in Series 2 - Explains the double bonded oxygen is a strong dipole - Explains H-bonding, and identifies the relative strength of H-bonding	3
- Identifies dispersion force and H-bonding in Series 2 / dispersion force and dipole for Series 1 OR - identifies one force and explains the trend (length and number of dispersion forces)	2
Identifies dispersion force OR H-bonding in Series 2 OR dipole for Series 1	1

Sample Answer:

Increase in the boiling point in both graphs:

- due to increase in chain length,
- increase the number on dispersion forces.
- C-2 of 20°C to C-3 of 49°C with the alkanal chain, and the same happened with the alcohol, with C-2 of 78°C to C-3 of 97°C.

Series 1:

- is a primary aldehyde/alkanal from the chemical formulae and being a homologous series.
- There is a dipole attraction (double bonded oxygen).

Series 2:

- is an alcohol,
- contains –OH. This combination due to electronegativity allows for hydrogen bonds to form.
- This is lot stronger than a dipole which account for the difference in boiling point, Ethanal BP = 20°C but ethanol with the H-bonding has significantly higher BP = 78°C.

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Question 36 (8 marks)**Outcomes Assessed:** CH11/12-5, CH11/12-7, CH12-15**Targeted Performance Bands:** 2–6

Criteria	Marks
- Provides a clear judgment statement about the economic and environmental impact of an improved catalyst based on discussion points. - Demonstrates an extensive understanding of the chemical principles involved to explain how an improved catalyst reduces energy requirements and increases yield, including collision theory, factors that affect rate and equilibrium reactions - Communicates clearly, logically and concisely	8
- Provides a clear judgment statement about the economic and/or environmental impact of an improved catalyst based on discussion points - Demonstrates a thorough understanding of the chemical principles involved to explain how an improved catalyst reduces energy requirements and increases yield including collision theory, factors that affect rate and/or equilibrium reactions - Communicates clearly and logically	6–7
- Demonstrates a sound understanding of the chemical principles involved to explain how an improved catalyst reduces energy requirements and/or increases yield - Discusses the economic and/or environmental impact of an improved catalyst	4–5
Demonstrates some understanding of a relevant chemical principle involved and/or outlines the economic or environmental impact	2–3
Provides some relevant information	1

Sample Answer:

Developing a catalyst that reduces the operational temperature from 450°C to a much lower temperature will have both a positive environmental and economic impact.

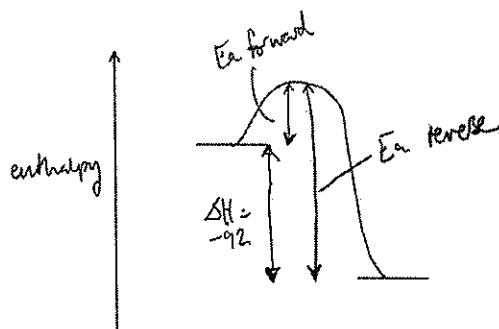
Reaction Kinetics:

For a reaction to proceed, reacting particles must collide with enough kinetic energy and the correct orientation. The energy required is called the activation energy (E_a) and raising the average energy of the particles by increasing temperature speeds up the reaction by increasing the frequency of successful collisions. Industrially, it is important to have a high reaction rate to make more product in a shorter timeframe to sell and this can be achieved in a gaseous system by raising the temperature and pressure. However, these operational conditions require a lot of energy and are therefore expensive to achieve.

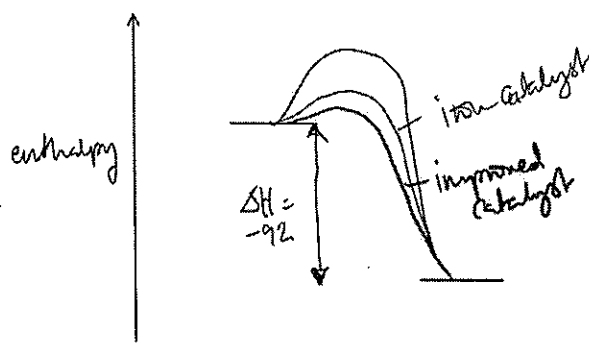
The Haber-Bosch process (HBP) is a reversible reaction that is exothermic in the forward direction. The forward reaction has a higher E_a than the reverse reaction as shown in this diagram:

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A catalyst works by providing an alternative reaction pathway with a lower E_a so collisions between particles with less kinetic energy will still react and the temperature doesn't need to be as high. For a reversible reaction, the E_a for both the forward and reverse reaction are reduced by the same amount.



However, even with an iron catalyst, the temperature has to be raised to 450°C to increase the production rate. An improved catalyst will further reduce the E_a so the reaction can operate at a high rate at a temperature much lower than 450°C . Reducing the energy required for heating lowers operational costs for the ammonia manufacturer and should reduce the cost to buyers, thus having an economic benefit. If the electricity is generated from fossil fuels, reducing energy demands will lead to a decrease in carbon dioxide emissions, reducing the contribution of ammonia production to climate change.

Equilibrium Considerations

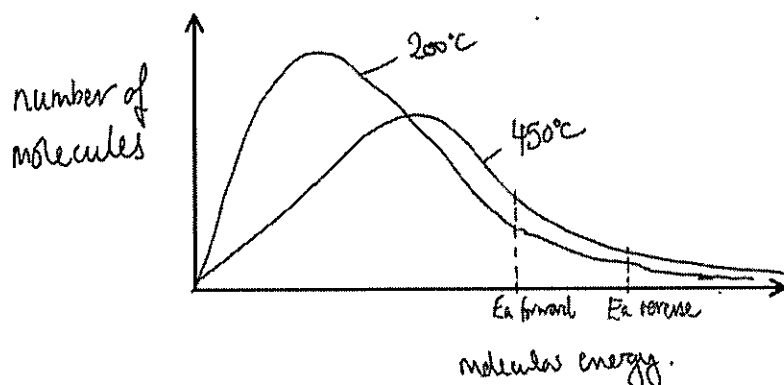
Since the forward reaction is exothermic, operating at lower temperatures will lead to a greater yield of ammonia as the forward reaction is favoured. The equilibrium constant (K_{eq}) will increase as temperature decreases; there will be a higher concentration of ammonia and lower concentrations of nitrogen and hydrogen in the equilibrium mixture:

$$K_{eq} = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$$

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At lower temperatures, the proportion of molecules that exceed the E_a for the reverse reaction is significantly lower than the E_a for the forward reaction. The area under the Maxwell-Boltzmann distribution graph shows the total number of particles with greater than the E_a for the forward and reverse reactions at 450°C and 200°C. The decrease in temperature halves the number of particles with enough energy for the reverse reaction, whereas it only reduces the number of particles with enough energy for the forward reaction by about 1/3, so the forward reaction is favoured, producing more ammonia.



With an iron catalyst, the temperature has to quite high to make ammonia in a shorter time frame even though total yield of ammonia is lower. With an improved catalyst, lower temperatures will be possible and the total yield of ammonia will be higher. This will increase the profits for the manufacturer as they have more ammonia to sell.

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